

Assignment 06: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1
library(tidyverse)
library(here)
library(rvest)
library(ggplot2)

here()
```

```
## [1] "/home/guest/872L/EDE_Spring2026"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Set the year to 2024
 - Scroll down and select the LWSP link next to Durham Municipality.
 - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwdid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

#2

```
Durham_WaterSupply_Webpage <- read_html(  
  'https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-32-010&year=2024')  
  
Durham_WaterSupply_Webpage  
  
## {html_document}  
## <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">  
## [1] <head>\n<title>DWR :: Local Water Supply Planning</title>\n<meta http-equ ...  
## [2] <body id="plan">\r\n<!--<div id="division-header">\r\n<a name="top" href= ...
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
 - Water system name
 - Contact Person
 - Ownership
- From the “3. Water Supply Sources” section:
 - Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “Reginald Hicks”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

#3

```
water_system <- Durham_WaterSupply_Webpage %>%  
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%  
  html_text()  
water_system
```

```
## [1] "Durham"
```

```
contact <- Durham_WaterSupply_Webpage %>%  
  html_nodes("div+ table tr:nth-child(4) td:nth-child(2)") %>%  
  html_text()  
contact
```

```
## [1] "Reginald Hicks"
```

```
ownership <- Durham_WaterSupply_Webpage %>%  
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%  
  html_text()  
ownership
```

```
## [1] "Municipality"
```

```
maximum_daily_use <- Durham_WaterSupply_Webpage %>%
  html_nodes("th~ td+ td") %>%
  html_text()
maximum_daily_use
```

```
## [1] "34.5000" "36.0600" "37.3300" "32.1000" "46.6500" "37.3600" "38.2000"
## [8] "41.9000" "36.5800" "36.7300" "42.9600" "34.4500"
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```
#4
daily_use_2024 <- data.frame(
  "Month" = c(
    "Jan", "May", "Sep", "Feb", "Jun", "Oct", "Mar", "Jul", "Nov", "Apr", "Aug", "Dec"),
  "Year" = rep(2024, 12),
  "Max_Daily_Use_mgd" = as.numeric(maximum_daily_use),
  stringsAsFactors = TRUE
)

daily_use_2024 <- daily_use_2024 %>%
  mutate(Water_System_Name = !!water_system,
         Contact_Name = !!contact,
         Type_of_Ownership = !!ownership,
         Date = my(paste(Month, "/", Year)))

daily_use_2024
```

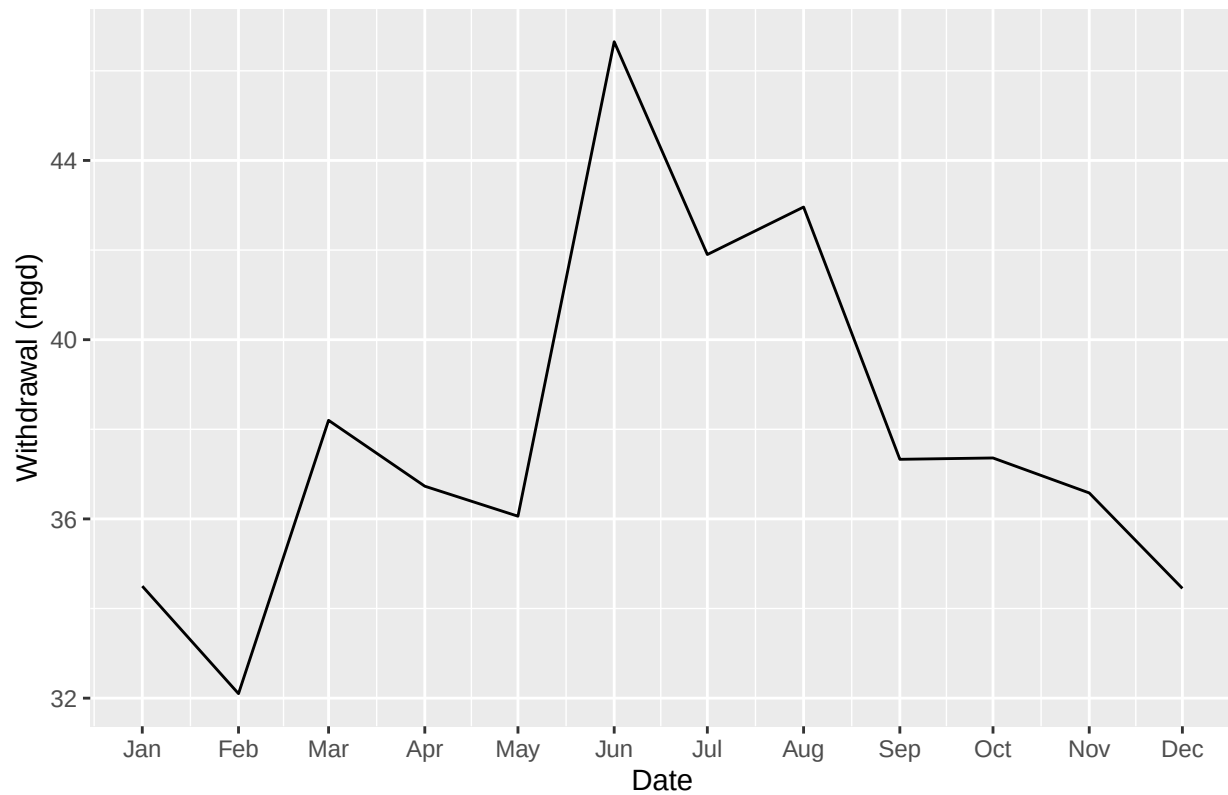
```
##   Month Year Max_Daily_Use_mgd Water_System_Name Contact_Name
## 1   Jan 2024          34.50      Durham Reginald Hicks
## 2   May 2024          36.06      Durham Reginald Hicks
## 3   Sep 2024          37.33      Durham Reginald Hicks
## 4   Feb 2024          32.10      Durham Reginald Hicks
## 5   Jun 2024          46.65      Durham Reginald Hicks
## 6   Oct 2024          37.36      Durham Reginald Hicks
## 7   Mar 2024          38.20      Durham Reginald Hicks
## 8   Jul 2024          41.90      Durham Reginald Hicks
## 9   Nov 2024          36.58      Durham Reginald Hicks
```

## 10	Apr 2024	36.73	Durham Reginald Hicks
## 11	Aug 2024	42.96	Durham Reginald Hicks
## 12	Dec 2024	34.45	Durham Reginald Hicks
##	Type_of_Ownership	Date	
## 1	Municipality	2024-01-01	
## 2	Municipality	2024-05-01	
## 3	Municipality	2024-09-01	
## 4	Municipality	2024-02-01	
## 5	Municipality	2024-06-01	
## 6	Municipality	2024-10-01	
## 7	Municipality	2024-03-01	
## 8	Municipality	2024-07-01	
## 9	Municipality	2024-11-01	
## 10	Municipality	2024-04-01	
## 11	Municipality	2024-08-01	
## 12	Municipality	2024-12-01	

#5

```
ggplot(daily_use_2024,aes(x=Date,y=Max_Daily_Use_mgd)) +
  geom_line() +
  scale_x_date(date_breaks = "1 month", date_labels = "%b") +
  labs(title = paste("2024 Water Usage Data for",water_system),
       y="Withdrawal (mgd)",
       x="Date")
```

2024 Water Usage Data for Durham



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs, “PWSID” and “year”, that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

```
#6.
Scraping.Water.Supply <- function(the_year, PWSID){

  water_supply_website <- read_html(
    paste0(
      "https://www.ncwater.org/WUDC/app/LWSP/report.php?",
      "pwsid=", PWSID,
      "&year=", the_year))

  water_system_tag <- "div+ table tr:nth-child(1) td:nth-child(2)"
  ownership_tag <- "div+ table tr:nth-child(2) td:nth-child(4)"
  contact_tag <- "div+ table tr:nth-child(4) td:nth-child(2)"
  max_daily_use_tag <- "th~ td+ td"

  the_water_system <- water_supply_website %>%
    html_nodes(water_system_tag) %>%
    html_text()
  the_ownership <- water_supply_website %>%
    html_nodes(ownership_tag) %>%
    html_text()
  the_contact_person <- water_supply_website %>%
    html_nodes(contact_tag) %>%
    html_text()
  the_max_daily_use_mgd <- water_supply_website %>%
    html_nodes(max_daily_use_tag) %>%
    html_text()

  max_daily_use_df <- data.frame(
    "Month" = c("Jan", "May", "Sep", "Feb", "Jun", "Oct", "Mar",
                "Jul", "Nov", "Apr", "Aug", "Dec"),
    "Year" = rep(the_year, 12),
    "PWSID" = rep(PWSID, 12),
    "Max_Daily_Use_mgd" = as.numeric(the_max_daily_use_mgd),
    stringsAsFactors = TRUE
  ) %>%
  mutate(Water_System_Name = !!the_water_system,
         Contact_Name = !!the_contact_person,
         Type_of_Ownership = !!the_ownership,
         Date = my(paste(Month, "/", Year)))

  #Return the dataframe
  return(max_daily_use_df)
}
```

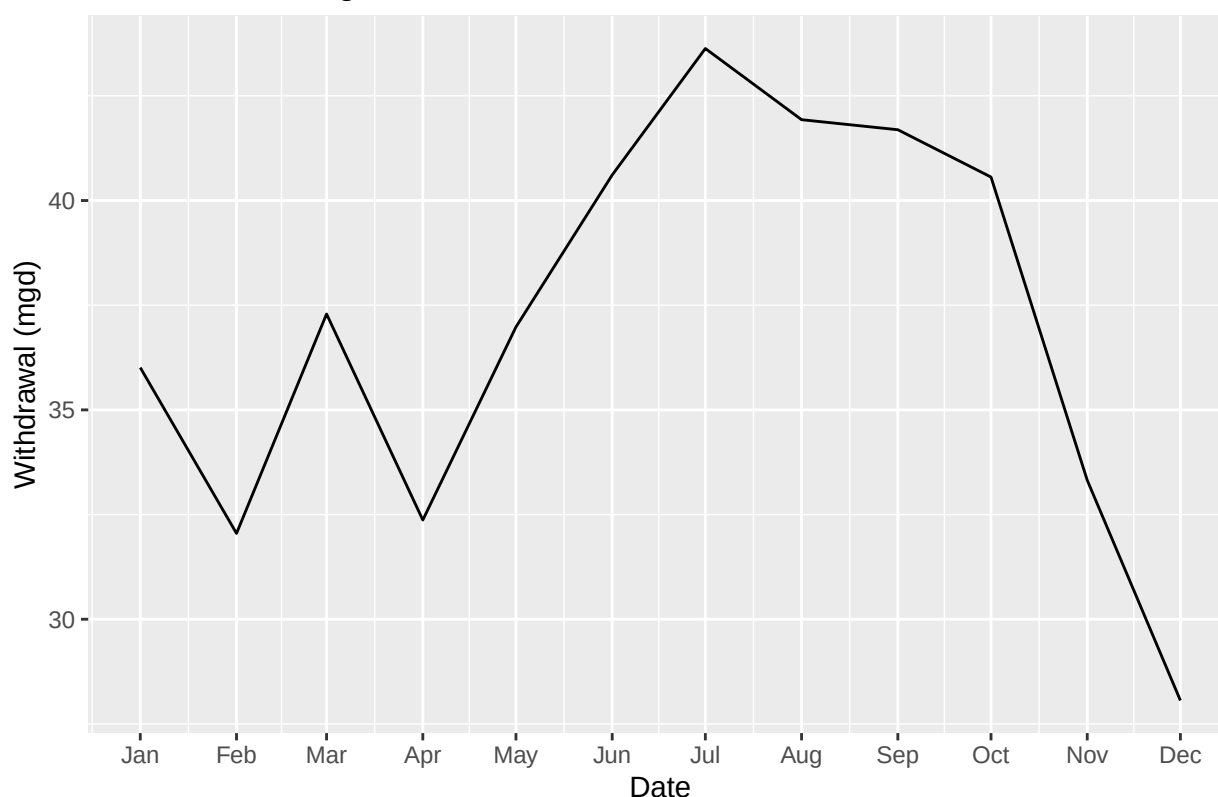
7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID=‘03-32-010’)

for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```
#7
Durham_2020_df <- Scraping.Water.Supply(2020,"03-32-010")
view(Durham_2020_df)

ggplot(Durham_2020_df,aes(x=Date,y=Max_Daily_Use_mgd)) +
  geom_line() +
  scale_x_date(date_breaks = "1 month", date_labels = "%b") +
  labs(title = paste("2020 Water Usage Data for Durham"),
       y="Withdrawal (mgd)",
       x="Date")
```

2020 Water Usage Data for Durham



```
str(Durham_2020_df)
```

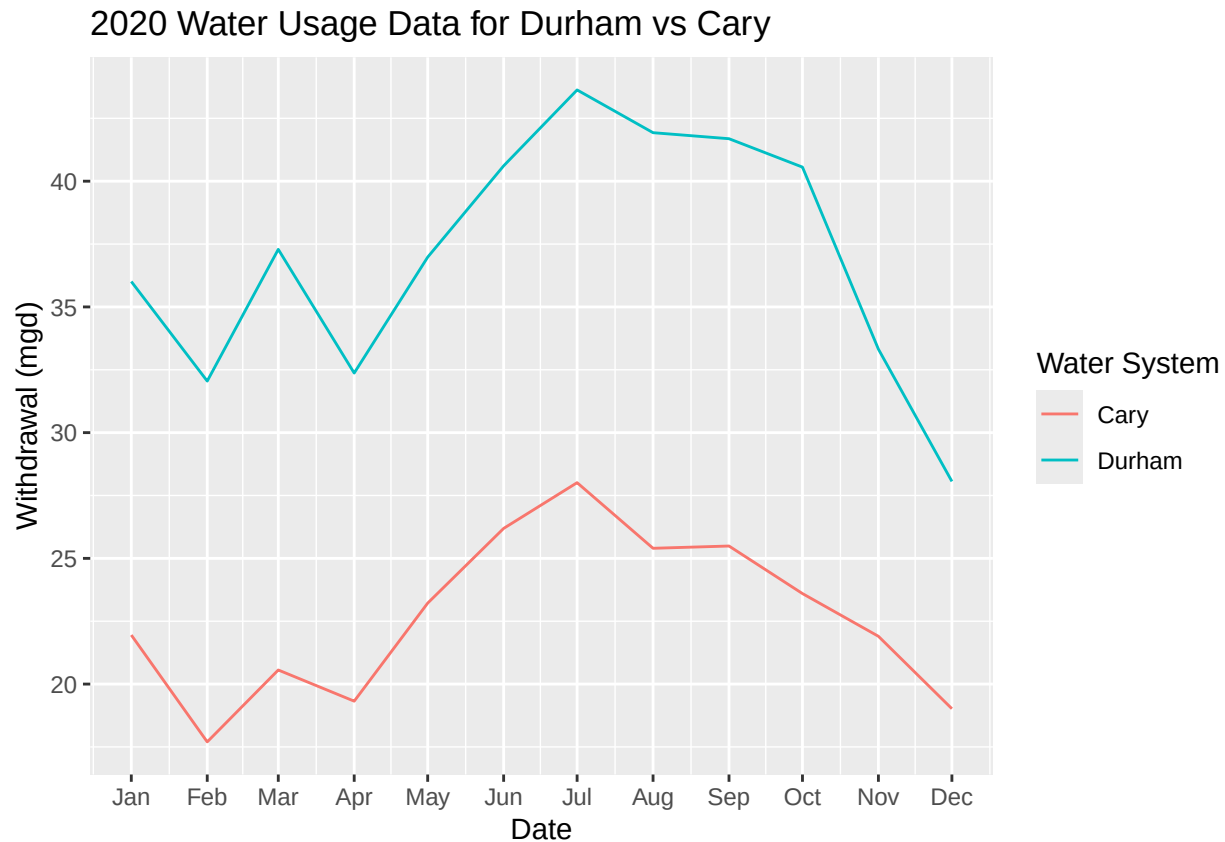
```
## 'data.frame': 12 obs. of 8 variables:
## $ Month : Factor w/ 12 levels "Apr","Aug","Dec",...: 5 9 12 4 7 11 8 6 10 1 ...
## $ Year : num 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ PWSID : Factor w/ 1 level "03-32-010": 1 1 1 1 1 1 1 1 1 1 1 ...
## $ Max_Daily_Use_mgd: num 36 37 41.7 32 40.6 ...
## $ Water_System_Name: chr "Durham" "Durham" "Durham" "Durham" ...
## $ Contact_Name : chr "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" ...
## $ Type_of_Ownership: chr "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ Date : Date, format: "2020-01-01" "2020-05-01" ...
```

8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8
Cary_2020_df <- Scraping.Water.Supply(2020,"03-92-020")
view(Cary_2020_df)

Combined_df <- bind_rows(Cary_2020_df,Durham_2020_df)

ggplot(Combined_df,aes(x=Date,y=Max_Daily_Use_mgd,colour = Water_System_Name)) +
  geom_line() +
  scale_x_date(date_breaks = "1 month", date_labels = "%b") +
  labs(title = paste("2020 Water Usage Data for Durham vs Cary"),
       y="Withdrawal (mgd)",
       x="Date",
       colour="Water System")
```



9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "06_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs (where one of the two inputs does not change). Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

#9

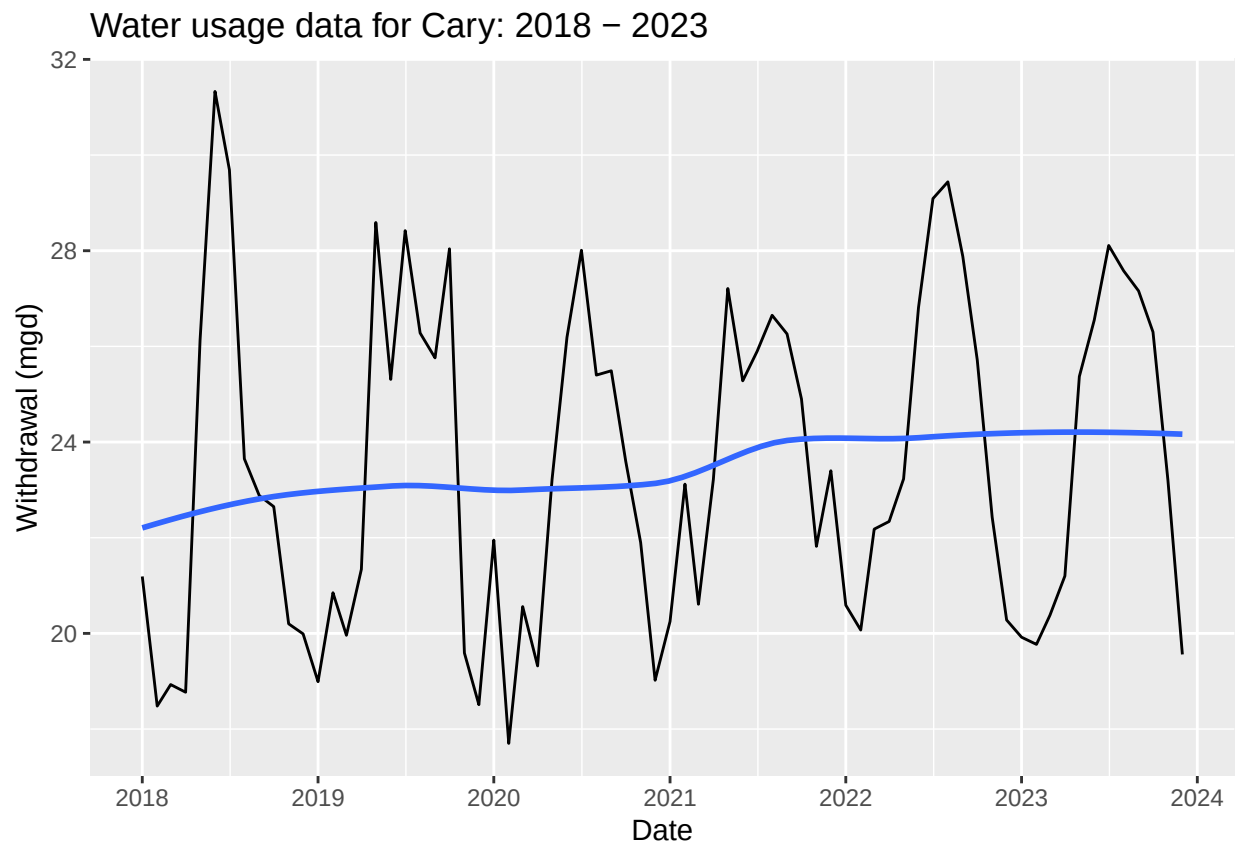
```
multiple_years = rep(2018:2023)
cary_pwsid = "03-92-020"

cary_1823 <- map(multiple_years, Scraping.Water.Supply, PWSID=cary_pwsid)

cary_combined_df <- bind_rows(cary_1823)

ggplot(cary_combined_df, aes(x=Date, y=Max_Daily_Use_mgd)) +
  geom_line() +
  geom_smooth(method="loess", se=FALSE) +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  labs(
    title = paste(
      "Water usage data for Cary:",
      multiple_years[1], "-", multiple_years[length(multiple_years)]
    ),
    y="Withdrawal (mgd)",
    x="Date"
  )
```

'geom_smooth()' using formula = 'y ~ x'



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: Water usage in Cary seems to be fluctuating consistently with peaks reached between

years followed by sudden drops. However, the regression line in the graph shows an overall linear trend, almost evening out these fluctuations. It seems like overall withdrawal has been increasing slightly with an average of 22mgd of withdrawal in the beginning of data collection and an average of 24mgd of withdrawal towards the end of data collection.